

The living or dead state of the cells in the next step of the simulation depends entirely on their current state. The cells don't "remember" any older states. There is a large body of research regarding the patterns that these simple rules produce. Tragically, Professor Conway passed away of complications from COVID-19 in April 2020. More information about Conway's Game of Life can be found at https://en.wikipedia.org/wiki/Conway%27s_Game_of_Life, and more information about Martin Gardner at https://en.wikipedia.org/wiki/Martin_Gardner.

The Program in Action

When you run `conwaysgameoflife.py`, the output will look like this:

```

      0          0          00      0 0
0    0  0  0    0          0          0 0000      0 00
00   0  0    0          0          0          0 0
00      0  0          00          00          00
00      00          0 0  0          00          00
          000          00  00          0 00
          00          0  000          0
          0          0  00          0 0
          00 00          00  0
          000          0000  0 0
          0  00          0 00 00 0  0  00
          0 0          0  0          0 00 0  0 000
          0          0000 00          00  0  00000 0
00      0          0  000  0 000          0000      0

```

How It Works

The state of the cells is stored in dictionaries in the `cells` and `nextCells` variables. Both dictionaries have (x, y) tuples for keys (where x and y are integers), '0' for living cells, and ' ' for dead cells. Lines 40 to 44 are set up to print a representation of these dictionaries onto the screen. The `cells` variable's dictionary represents the current state of the cells, while `nextCells` stores the dictionary for the cells in the next step in the simulation.

```

1. """Conway's Game of Life, by Al Sweigart al@inventwithpython.com
2. The classic cellular automata simulation. Press Ctrl-C to stop.
3. More info at: https://en.wikipedia.org/wiki/Conway%27s_Game_of_Life
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: short, artistic, simulation"""
6.
7. import copy, random, sys, time
8.
9. # Set up the constants:
10. WIDTH = 79 # The width of the cell grid.
11. HEIGHT = 20 # The height of the cell grid.
12.
13. # (!) Try changing ALIVE to '#' or another character:

```